

Design Proposal: Waveguide, Y Branch and Mach-Zehnder Interferometer

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Abstract—This proposal outlines the design and simulation of a compact Mach-Zehnder Interferometer (MZI) using silicon-on-insulator (SOI) strip waveguides. The design targets photonic integrated circuits (PICs). There are 4 wave guide variants with length differences, 4 single-output MZI with length differences and 2 dual-output MZI variants with path-length differences (ΔL) that are simulated using Lumerical MODE and layouted using KLayout. The designs are validated by automated layout verification in KLayout. The proposed design adheres to foundry-compatible design rules for automated testing.

Index Terms—photonic, waveguide, FSR, group index, MZI

I. INTRODUCTION

Silicon photonics has emerged as a dominant platform for photonic-electronic integration, driven by CMOS-compatible fabrication and cost-effective scalability [1]. A key component in this ecosystem is the Mach-Zehnder Interferometer (MZI), widely used for wavelength filtering, modulation, and sensing [2]. This work addresses the need for compact, fabrication-tolerant MZI designs optimized for the C-band (1550 nm).

This work focuses on the design and simulation of a compact MZI using silicon-on-insulator (SOI) strip waveguides, optimized for operation in the C-band (1550 nm), which is the standard wavelength range for telecommunications. The proposed design includes four waveguide variants, four Single-Output MZI with length differences, and two MZI variants with path-length differences (ΔL). These designs are simulated using Lumerical MODE to analyze their optical performance and laid out using KLayout to ensure compliance with foundry-compatible design rules. The simulations aim to validate the performance of the MZI in terms of insertion loss, free spectral range (FSR), and fabrication tolerance, while the layout process ensures that the designs are ready for automated testing and fabrication.

II. THEORY

A. Waveguide Fundamentals

Silicon-on-insulator (SOI) strip waveguides are the backbone of integrated photonic circuits. They consist of a silicon core ($n_{\text{Si}} \approx 3.47$) surrounded by a buried oxide (SiO_2 , $n_{\text{SiO}_2} \approx 1.44$) and cladding (air or oxide). Light is confined in the high-index silicon core via total internal reflection.

B. Y-Branch Splitter/Combiner

The Y-branch is a passive optical component used to split or combine light between two waveguides. It operates by adiabatically dividing an input optical signal into two arms or combining two input signals into a single output. When light enters the input waveguide of a Y-branch splitter, it is gradually divided into two separate waveguides. Conversely, in a Y-branch combiner, two input signals interfere coherently at the output waveguide. For a balanced Y-branch, the power splitting ratio is ideally equal, meaning each output arm carries half of the input power, expressed as $P_1 = P_2 = P_{\text{in}}/2$.

However, practical Y-branches exhibit insertion loss due to splitting and mode mismatch. In SOI-based Y-branches, typical insertion losses are less than 0.2 dB. To minimize losses, the design incorporates adiabatic tapering, which ensures a slow transition between the input and output waveguides, and S-bend geometries, which reduce radiation loss at the split.

C. Mach-Zehnder Interferometer (MZI)

The Mach-Zehnder Interferometer (MZI) is a key component in photonic circuits, used to measure phase differences between two optical paths. It consists of two Y-branches: one acting as a splitter at the input and the other as a combiner at the output. Between these Y-branches are two waveguide arms, one of which has a path-length difference ΔL relative to the other. This path-length difference introduces a phase shift $\Delta\phi$, given by

$$\Delta\phi = \beta\Delta L = \frac{2\pi n_{\text{eff}}\Delta L}{\lambda}$$

where β is the propagation constant, n_{eff} is the effective index, and λ is the wavelength of light. The output intensity of the MZI depends on the phase difference between the two arms and is expressed as $I_{out} = 2I_{in}[1 + \cos(\Delta\phi)]$.

This results in a periodic transmission spectrum, with peaks corresponding to constructive interference and troughs to destructive interference. The spacing between adjacent peaks, known as the free spectral range (FSR), is given by $FSR = \lambda^2 / ng\Delta L$. [1]

From the discussion of last session, we can obtain the transfer function of MZI

$$T_{MZI}(\lambda) = (\frac{1}{2}) [1 + \cos(\beta\Delta L)]$$

The free spectral range (FSR) of the interferometer can be written as $FSR = \lambda^2 / (ng\Delta L)$. The FSR with different ΔL in our design is shown in Table 1.

$\Delta L(\mu m)$	FSR(nm)
130	~4.3
387	1.32

Table 1

C.1 Single-Output Mach-Zehnder Interferometer (MZI)

The Single-Output Mach-Zehnder Interferometer (MZI) is a passive optical component used to recombine light from two optical paths into a single output. It operates by splitting an input optical signal into two arms, allowing independent phase accumulation, and then recombining them into a single output waveguide without further splitting.

When light enters the input waveguide, it is adiabatically split into two arms. After traveling through paths of possibly different lengths, the two optical signals interfere coherently at the output. Ideally, if no phase difference exists between the two arms, full constructive interference occurs at the output port.

Insertion loss is an important practical consideration, arising from imperfections in splitting and mode mismatch during recombination. In silicon-on-insulator (SOI) platforms, careful use of adiabatic tapering and optimized S-bend geometries can reduce insertion losses to below 0.2 dB, minimizing scattering and radiation losses.

C.2 Dual-Output Mach-Zehnder Interferometer (MZI)

The Dual-Output Mach-Zehnder Interferometer is a key component in integrated photonics, widely used to measure phase differences and perform optical signal processing. It consists of two 50:50 splitters (e.g., directional couplers or multimode interference couplers), one at the input and one at the output, connecting two parallel waveguide arms.

differences ΔL in the design result in different FSR values, as summarized in Table 1.

III. MODELING AND SIMULATION

A. Waveguide

In our design, we use strip waveguide with 500nm width and 220nm height. The simulated waveguide mode profiles are presented in the following figures.

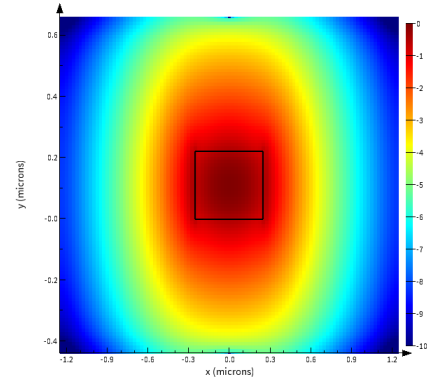


Fig. 1. Electric Field Intensity of the first TE Mode Log Scale

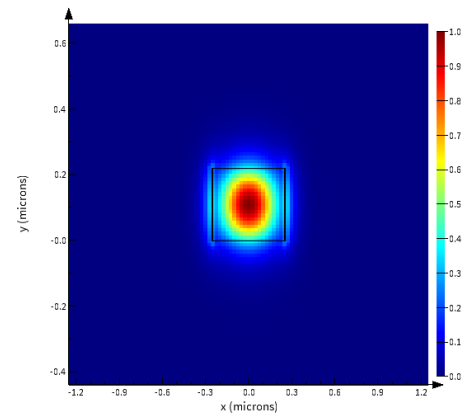


Fig. 2. Electric Field Intensity of the first TE Mode Linear Scale

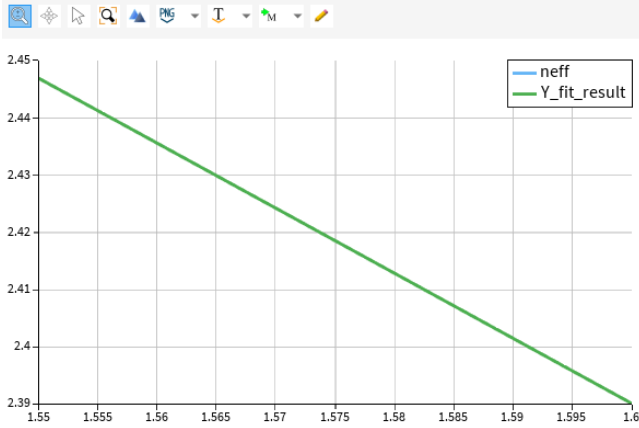


Fig. 3. Effective index vs wavelength

From figures above We can see that the effectiveness index decreases as the wavelength increases, while the group and the intensity output can be represented as index increases with the increase in wavelength. The compact model calculation formula is

$$n_{eff}(\lambda) = 2.44 - 1.13(\lambda - \lambda_0) - 0.03(\lambda - \lambda_0)^2$$

B. Y Branch

These are the simulation result for Y Branch using numerical MODE

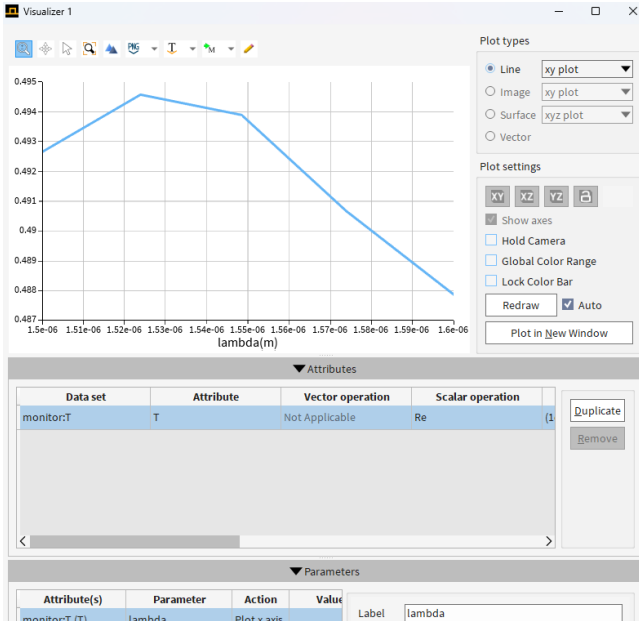


Fig. 4. Y-Branch simulation using Lumerical MODE

C. Chip Layout Design

The proposed design includes four waveguide variants, two Single-Output MZI with length differences, and two MZI variants with path-length differences (ΔL). These designs are simulated using Lumerical MODE to analyze their optical

performance and laid out using KLayout to ensure compliance with foundry-compatible design rules.

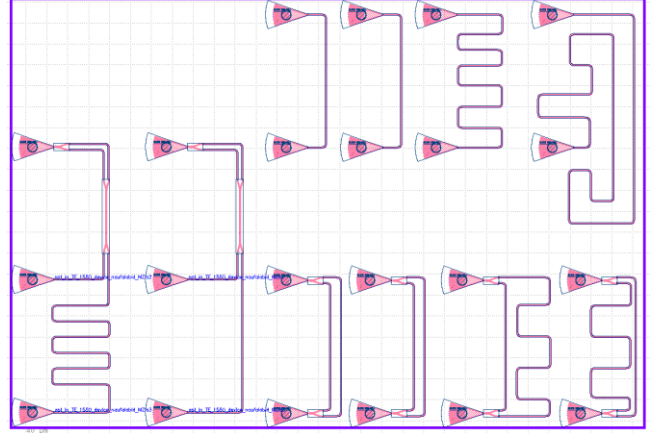


Fig. 5. Proposed Layout

IV. FABRICATION PROCESS

A. Fabrication Process

Fabrication was performed at one or more of the following facilities: Applied Nanotools and the Washington Nanofabrication Facility. The processes used are described below.

At Applied Nanotools, Inc., the photonic devices were fabricated using the NanoSOI MPW process, which is based on direct-write 100 keV electron beam lithography. The process utilized 200 mm diameter silicon-on-insulator wafers with a 220 nm device layer and a 2 μ m buried oxide. Wafers were pre-diced into 25 $\times$ 25 mm substrates, and scribe lines were added to the backsides to facilitate separation into smaller chips after fabrication. The substrates underwent an initial piranha clean (3:1 $H_2SO_4:H_2O_2$) for 15 minutes, followed by a rinse in water and isopropanol. Hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrates and baked to remove the solvent. Patterning was carried out using a JEOL JBX-8100FS electron beam lithography system at the University of British Columbia. Proximity effect correction was applied, and shape writing order was optimized to reduce beam drift and improve pattern fidelity.

Following exposure and development in tetramethylammonium hydroxide (TMAH), the devices were optically inspected for residues and defects. The chips were then mounted on a 4-inch handle wafer and etched using an anisotropic ICP-RIE process with chlorine, after qualification of the etch rate. Residual resist was removed using a 10:1 buffered oxide etch, and scanning electron microscopy (SEM) was used to verify pattern and etch quality. A 2.2 μ m oxide cladding was deposited using plasma-enhanced chemical

vapor deposition (PECVD) with tetraethyl orthosilicate (TEOS) at 300 °C. Reflectometry measurements were taken throughout the process to confirm the thicknesses of the device layer, buffer oxide, and cladding.

At the Washington Nanofabrication Facility, devices were fabricated using 100 keV electron beam lithography. The process used silicon-on-insulator wafers with a 220 nm silicon layer on a 3 μm silicon dioxide layer. The substrates, 25 mm squares diced from 150 mm wafers, were cleaned with a solvent rinse followed by a hotplate dehydration bake. HSQ resist (Dow-Corning XP-1541-006) was spin-coated at 4000 rpm and baked at 80 °C for 4 minutes.

Patterning was performed using a JEOL JBX-6300FS system operated at 100 keV, with an 8 nA beam current and 500 μm exposure field size. The shape placement grid was set to 1 nm, and the beam stepping grid was 6 nm. An exposure dose of 2800 $\mu\text{C}/\text{cm}^2$ was used. After exposure, the resist was developed in 25% TMAH for 4 minutes, followed by rinsing with deionized water for 60 seconds, isopropanol for 10 seconds, and drying with nitrogen. Unexposed silicon was etched using inductively coupled plasma etching in an Oxford Plasmalab System 100 with a chlorine gas flow of 20 sccm, pressure of 12 mTorr, ICP power of 800 W, bias power of 40 W, and platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using PECVD in the same system with a silane (SiH_4) flow of 13.0 sccm, nitrous oxide (N_2O) flow of 1000.0 sccm, nitrogen (N_2) flow of 500.0 sccm, pressure at 1400 mTorr, RF power of 120 W, and a platen temperature of 350 °C. During deposition, the chips were placed directly on a silicon carrier wafer and buffered on all sides with silicon pieces to improve uniformity.

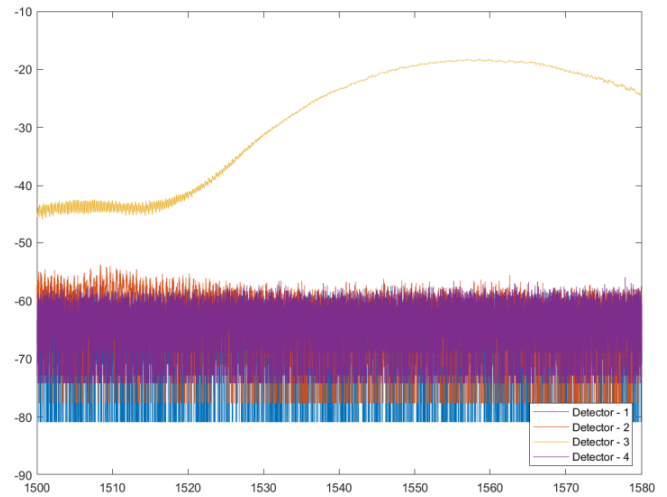
B. Manufacturing Challenges

Due to fabrication issues, there are several aspects that include mismatches between the designed and the fabricated devices. Mainly, these manufacturing errors generate variabilities mainly on thickness and width of the waveguides. As an approximation, it is usual to associate the fabrication variability of a given geometrical parameter with a Gaussian distribution, where the nominal value corresponds to the distribution mean, and the fabrication error is expressed by the standard deviation σ .

V. EXPERIMENTAL RESULTS

At the time of writing, simulation results have been completed for the waveguide and single-output MZI designs. Full simulation for the dual-output MZI structures is still pending. All layouts were verified and passed foundry design rule checks.

A. Waveguide Results

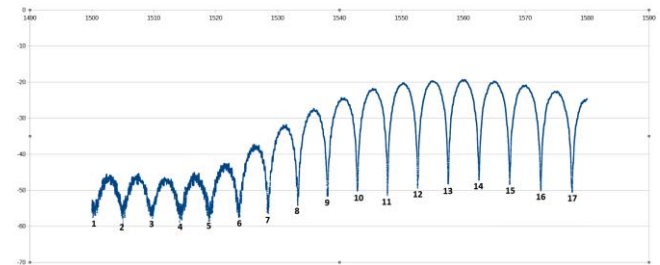


Simulations using Lumerical MODE confirm that the 500 nm $\times$ 220 nm strip waveguides support a well-confined fundamental TE mode. The effective index (n_{eff}) decreases with increasing wavelength, consistent with typical dispersion behavior in SOI platforms. Electric field mode profiles show strong confinement with low expected propagation loss.

B. Single-Output MZI

These devices are Mach-Zehnder Interferometers with one output port, where two arms recombine light without further splitting. Simulations show near-equal power interference at the output, confirming effective recombination and low insertion loss. The design uses adiabatic tapers and smooth transitions to minimize scattering and reflection losses.

One of my single output MZI called yD have this measurement result:



The peaks of the interference are located on this location:

	nm	dB
1	1500.59	-56,7
2	1505.01	-57.3
3	1509.79	-56.9
4	1514.54	-58.2
5	1519.08	-56.5
6	1523.89	-56.9
7	1528.5	-55.7

To find the group index we need to calculate the FSR. Which can be calculated by this:

- $1505.01 - 1500.59 = 4.42 \text{ nm}$
- $1509.79 - 1505.01 = 4.78 \text{ nm}$
- $1514.54 - 1509.79 = 4.75 \text{ nm}$
- $1519.08 - 1514.54 = 4.54 \text{ nm}$
- $1523.89 - 1519.08 = 4.81 \text{ nm}$
- $1528.50 - 1523.89 = 4.61 \text{ nm}$
- $1533.25 - 1528.50 = 4.75 \text{ nm}$

$$\begin{aligned} \text{Average FSR} &= 4.42 + 4.78 + 4.75 + 4.54 + 4.81 + 4.61 \\ &\quad + 4.75 \\ &= 32.66 \text{ nm} \end{aligned}$$

$$\text{Average FSR} = \frac{32.66}{7} \approx 4.66 \text{ nm}$$

With:

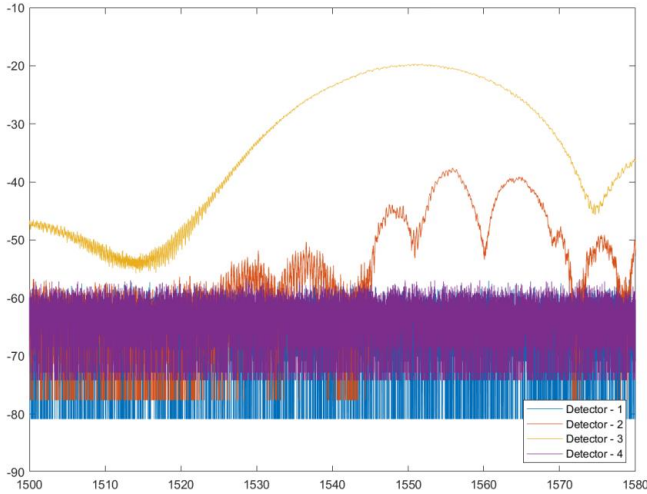
- $\Delta L = 117.386 \text{ } \mu\text{m} = 1.17386 \times 10^{-4} \text{ m}$
- $\lambda = 1540 \text{ nm} = 1.54 \times 10^{-6} \text{ m}$
- $\text{FSR} = 4.66 \text{ nm} = 4.66 \times 10^{-9} \text{ m}$

Using the group index formula:

$$ng = \frac{\lambda^2}{\Delta L \times \text{FSR}} \approx 4.33$$

These are very close to the simulation result from table 1. Which mean the chip result are correct.

C. Dual-Output MZI



These are conventional MZIs with both arms recombined into two output ports, allowing observation of full interference fringes. While the designs and layouts have been completed, full optical simulation results (e.g., spectral transmission and free spectral range) are still in progress. Theoretical FSR values are shown in Table 1.

VI. CONCLUSION

In this article, we propose 2 double output MZI with different ΔL , 4 waveguide and 4 single output MZI using silicon strip waveguide. The simulations of waveguides and MZI have been given in the paper. The simulation results coincide with our expectation.

REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge University Press, 2015.
- [2] J. Zhang et al., "Ultra-Compact Mach-Zehnder Interferometer in SOI," *Photonics*, vol. 8, no. 10, 2021.